Darshit Joshi

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Summary

Senior Software Engineer with 3.5+ years of experience in AI, Data science and backend applications building production agentic AI systems, LLM/GenAI applications and large scale data and ML platforms. Co-architected a multi-tenant AI agent platform serving 50+ production AI capabilities over MCP tools, plus multi-agent GenAI pipelines that cut manual effort 80% and cost 60%. Depth in multi-agent orchestration, RAG and vector search, prompt and context engineering, computer vision, distributed pipelines and Python, Go, FastAPI, MLOps/LLMOps.

Skills

Agentic AI & GenAI: AI Agents, Multi-Agent Systems, LangChain, LangGraph, Strands Agents, RAG, Tool/Function Calling, Prompt & Context Engineering, Human-in-the-Loop, Voice AI (Nova Sonic, ElevenLabs, Twilio), Cost & Latency Optimization
ML & Generative Models: PyTorch, XGBoost, Recommender & Ranking Systems, Bayesian Optimization (Optuna), MCMC, Hyperparameter Tuning, Stable Diffusion, ComfyUI, Qwen, LoRA & LLM Fine-Tuning, VAEs, CUDA, Speech/ASR
Data Engineering: ETL Pipelines, Control-M, Data Modeling, Query Optimization, SQL, PostgreSQL, MySQL, Snowflake, Sybase, MongoDB, Redis, OpenSearch, Vector Databases, PySpark, AWS Glue, SQLAlchemy, Pandas, Data Catalog Platforms
Development: Python, Java, C++, JavaScript, TypeScript, FastAPI, Django, SpringBoot, REST APIs, Microservices, Event-Driven Architecture, WebSockets, RabbitMQ, Celery, React, Next.js, System Design, Grafana, LogQL
Cloud, MLOps & DevOps: AWS (EC2, ECS Fargate, S3, Lambda, SQS, SageMaker, Bedrock, Transcribe, Connect, CloudWatch), Oracle Cloud, GCP (Vertex AI, Dataflow, Kubeflow), Docker, Kubernetes, ArgoCD, Jenkins, CI/CD, MLOps, Model Deployment & Serving, Grafana, LogQL
AI Tools: Claude Code, Codex, Antigravity, Cursor, Kiro, Copilot

Experience

Armakuni **Mar. 2025 – Present**

Sr. Software Engineer

- Served as **Solution Architect** and **Senior Software Engineer**, owning end-to-end system design for **4** client-facing production projects and leading delivery on **3**; shipped **6+ POCs** in 4–8 week cycles, mentored a 6-month intern cohort, and sat on hiring panels for campus and senior engineering recruitment.
- Designed and trained a CTR-optimization model for an ad agency using **XGBoost** and **MCMC**-based Bayesian inference, driving feature engineering and an end-to-end **MLOps pipeline** on **AWS SageMaker** for training, evaluation, and deployment.
- Built a real-time AI voice agent for a Florida-based real estate company handling **10K+ calls/month** and **500+ bookings/month** at **sub-500ms** response latency; automates appointment booking, availability checks, and live pricing via web scraping with guideline-driven case handling, powered by **AWS Bedrock**, **Amazon Nova Sonic**, **AWS Transcribe**, and **ElevenLabs** TTS.
- Built a GPU-accelerated generative image pipeline for a client on **Stable Diffusion** and **AWS EC2 GPU** instances, fine-tuning **4 custom LoRA** adapters with custom **VAEs** and checkpoint management; **CUDA**-optimized inference delivered **20+ images/min** at **70% lower cost**.
- Built an image-to-image generation system for a gaming company that preserves character identity across poses and frames, orchestrating **ComfyUI** workflows with **Qwen** image models on **AWS SageMaker** endpoints to generate **20+ consistent character poses/frames per minute**.
- Built a retrieval-augmented generation (**RAG**) patient-support assistant over **Twilio** SMS and voice, grounding **AWS Bedrock** LLM responses in an **OpenSearch** vector index; resolved **1K+ issues** with **85%** (850+) handled without human escalation at **<2s** response time, plus automated escalation, ticket tracking, and follow-ups.
- Engineered a **Bayesian optimization** framework with **Optuna** in **Python** on **AWS ECS** over **Snowflake** data, searching a trillion-plus space of detector-score threshold combinations; cached evaluations cut search time from years of brute force to **5–10 minutes** and produced business rules that **reduced false positives by 89%**.
- Built banking transaction-handling and data analytics pipelines for a bank client on **Oracle Cloud**, processing transaction data from **Sybase** and orchestrating batch workflows with **Control-M** to deliver reliable, scheduled analytics and reporting.

DosePack (Meditab) **Dec. 2022 – Mar. 2025**

Software Engineer

- Designed and shipped **FastAPI**-based **RESTful APIs** for on-premise pharmacy automation workflows, cutting response times by **50%** through SQL query optimization, caching, and **async I/O**, and standardizing service contracts for interoperability across internal systems and partner pharmacies.
- Built and published **dpws-migration-automation** on **PyPI**, automating version-controlled **PostgreSQL** schema migrations across dev, staging, and production via **CI** pipelines, cutting deployment time by **40%** and keeping schemas consistent across environments.
- Built **Python/Pandas** ETL pipelines scheduled via **cron** to ingest, cleanse, and transform pharmacy orders, robot telemetry, and pill-dispensing events, adding automated **validation** and **feature engineering**; automated daily performance reporting from **4 hours to 1 minute** for **7+ stakeholders** and **20+ senior leaders**.
- Engineered event-driven real-time features with **FastAPI WebSockets** and **Redis pub/sub**, streaming bi-directional updates across **7+ robots and devices**; built **DPNS**, a notification service for robot error alerts and fill-progress tracking, alongside live order status and dynamic operational dashboards.
- Designed and built a custom **HIPAA-compliant** auth service with **JWT** tokens, **RBAC**, and **OAuth 2.0** scope-based **SSO**, securing access across **5 microservices** and client-facing applications.
- Redesigned the **PostgreSQL** data model with schema normalization, **materialized views**, and a data archival strategy, delivering ~**60% faster** queries and improved storage utilization.
- Designed a constraint-aware **greedy algorithm** for pack-filling slot allocation in multi-dose medication packaging, replacing compute-intensive customization logic and improving robot fill-workflow throughput.
- Architected a rules-driven middleware service with **FastAPI**, **RabbitMQ**, and **Celery** that routes and balances **10K+ fill orders/day** across **4 US-based pharmacies**, delivering an **80% efficiency gain** and reducing robot idle time.

Achievements

- Training and Placement Coordinator at DDU (2023).
- Interviewer for campus placement drives and senior roles at Armakuni.
- Coordinator at DDU NSS club activities.

Education

Dharmsinh Desai University **Jul. 2019 – May. 2023**

B.Tech, CGPA: 8.72/10 *Nadiad, Gujarat*